

conditions, such as the consistently mild 25 degrees Celsius climate of Hawaii, these devices can achieve 50,000 to 100,000 life cycles. This remarkable durability suggests that hybrid supercapacitors could potentially last the entire lifespan of a residence—70 to 80 years—without needing replacement.

In comparison, mainstream battery systems like Tesla's Powerwall have specific throughput limits under warranty, typically around a 37-megawatt-hour total, which translates to about four and a half to five years of service assuming daily cycling. After this warranty period, these batteries tend to retain only about 70% of their initial capacity, indicative of a 30% loss due to degradation. This degradation primarily results from the Solid Electrolyte Interphase (SEI), a buildup that occurs in the anode and cathode and hinders electron and ion transfer, reducing battery capacity over time.

Hybrid supercapacitors, which merge the features of both capacitors and batteries, suffer less degradation because they rely partially on electrostatic mechanisms instead of purely chemical reactions. This results in them maintaining approximately 80 to 85% of their capacity over many years, which is a stark contrast to the 15 to 20% degradation typically seen in Tesla's products over a similar period.

These attributes make hybrid supercapacitors a superior choice for residential and commercial energy storage applications, offering longer life cycles, less degradation, and enhanced efficiency. Not only do they present a viable alternative for reducing electricity costs and boosting energy independence, but they are also environmentally friendly when it comes to disposal. Despite the recyclability of components such as copper, aluminium, and lithium, the extended lifespan of hybrid

supercapacitors means that the need for recycling has not yet become a practical concern, as most systems installed in the last decade continue to operate efficiently.

Bridging the gap between batteries and capacitors

The development of hybrid supercapacitors highlights ongoing innovation in energy storage technology. These devices merge the characteristics of traditional capacitors and batteries, enhanced through proprietary manufacturing processes that precisely apply materials to the anode and cathode and select the best electrolyte chemistry. These proprietary techniques, crucial for improving performance and longevity, are closely guarded as intellectual property by their manufacturers.

While pure lithium batteries like lithium nickel cobalt (LNC), lithium iron phosphate (LFP), and lithium titanate (LTO) offer high energy densities critical for electric vehicles (EVs), hybrid supercapacitors excel in providing robust, long-lasting energy storage. Lithium batteries are preferred in high-performance EVs like Tesla's Model S for maximising range and efficiency. However, these batteries still lag behind traditional fuels like gasoline in energy density, presenting a significant challenge in the automotive industry.

Advancements in lithium battery technology

Advancements in lithium battery technology, including the development of solid-state electrolytes, aim to meet consumer expectations for longer range, enhance safety features, and reduce degradation. These innovations not only improve battery performance but also benefit hybrid supercapacitors by allowing them to incorporate breakthroughs from the battery sector, enhancing their efficiency and functionality.

In the automotive industry, hybrid vehicles that combine efficient petrol engines with battery systems are making notable advancements. For example, BYD in China is producing a new hybrid vehicle with a range of up to 1931km (1200 miles) by integrating a modest battery capacity with a highly efficient petrol engine that recharges the battery while driving. This combination addresses range anxiety associated with fully electric vehicles by providing a petrol backup that significantly extends the range without frequent recharges. This evolving landscape indicates a growing demand for advanced energy storage solutions, spurring continued innovation and the integration of diverse technologies to create more efficient, sustainable, and versatile energy storage systems.

From stabilising the load on power grids during peak times to providing uninterrupted power during outages, hybrid supercapacitors offer a reliable solution that not only meets modern energy demands but does so in an environmentally conscious manner. As we continue to innovate and refine this technology, the potential for hybrid supercapacitors to revolutionise energy storage and management across the globe grows, promising a future where energy efficiency and reliability are paramount. **EFY**

Co-author Ashwini Kumar Sinha is a tech journalist at EFY, with hands-on expertise in electronics DIY. He has an extraordinary passion for AI, IoT, and electronics. He holds two design records and is two times winner of US-China Makers Award.

Co-author Nidhi Agarwal is a journalist at EFY. She is an Electronics and Communication Engineer with over five years of academic experience. Her expertise lies in working with development boards and IoT cloud. She enjoys writing as it enables her to share her knowledge and insights related to electronics with like-minded techies.